Topographic Radar Geometry-Guided Cross-Attention for Alpine Avalanche Debris Mapping in Sentinel-1 SAR: Cross-System Generalization Across the Alps, Pamirs, and Arctic

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Article Type: Original Research Article

Target Journal: Journal of Mountain Science (Springer Nature / Science Press, CAS)

Running Head: TopoRadar-Net for Sentinel-1 Avalanche Mapping

Word Count: ∼8,500 words (including text, equations, and references)

Number of Figures: 4 (vector diagrams and performance charts)

Number of Tables: 8 (cross-system benchmarks, statistical tests, ablations, stratification, triage matrix)

CRediT Authorship Contribution Statement:

- Akshay Sharma: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization.
- Lalji Prasad: Supervision, Methodology, Resources, Validation, Writing – Review & Editing.

Declaration of Competing Interest:
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding Information:
No external grant funding was received for this specific study.

Data and Code Availability:
The AvalCD benchmark dataset is publicly accessible on Zenodo (DOI: 10.5281/zenodo.15863589). All source code, neural network implementations, evaluation scripts, and replication suites are publicly accessible at https://github.com/akssha74/TopoRadar-Net-AvalCD. Model checkpoints for all three independent training seeds and publication artifacts are openly distributed under Release v1.0.0: https://github.com/akssha74/TopoRadar-Net-AvalCD/releases/tag/v1.0.0.